

$$99. \frac{x^2 + y^2}{x^2 - y^2} = \frac{\frac{x^2}{y^2} + 1}{\frac{x^2}{y^2} - 1} = \frac{\left(\frac{x}{y}\right)^2 + 1}{\left(\frac{x}{y}\right)^2 - 1} = \frac{\left(\frac{1}{3}\right)^2 + 1}{\left(\frac{1}{3}\right)^2 - 1}$$

$$= \frac{\left(\frac{1}{9} + 1\right)}{\left(\frac{1}{9} - 1\right)} = \frac{10}{9} \times \left(-\frac{9}{8}\right) = -\frac{5}{4}.$$

$$100. \frac{x}{2y} = \frac{6}{7} \Rightarrow \frac{x}{y} = \left(2 \times \frac{6}{7}\right) = \frac{12}{7}.$$

$$\therefore \frac{x-y}{x+y} + \frac{14}{19} = \frac{\frac{x}{y} - 1}{\frac{x}{y} + 1} + \frac{14}{19} = \frac{\frac{12}{7} - 1}{\frac{12}{7} + 1} + \frac{14}{19} = \frac{(5/7)}{(19/7)} + \frac{14}{19}$$

$$= \left(\frac{5}{7} \times \frac{7}{19}\right) + \frac{14}{19} = \frac{5}{19} + \frac{14}{19} = \frac{19}{19} = 1.$$

$$101. \frac{a}{b} = \frac{4}{5} \text{ and } \frac{b}{c} = \frac{15}{16} \Rightarrow \left(\frac{a}{b} \times \frac{b}{c}\right) = \left(\frac{4}{5} \times \frac{15}{16}\right) \Rightarrow \frac{a}{c} = \frac{3}{4}.$$

$$\therefore \frac{c^2 - a^2}{c^2 + a^2} = \frac{1 - \left(\frac{a^2}{c^2}\right)}{1 + \left(\frac{a^2}{c^2}\right)} = \frac{1 - \left(\frac{a}{c}\right)^2}{1 + \left(\frac{a}{c}\right)^2} = \frac{1 - \frac{9}{16}}{1 + \frac{9}{16}} = \frac{(7/16)}{(25/16)} = \frac{7}{25}.$$

$$102. x = y \Leftrightarrow 1 - q = 2q + 1 \Leftrightarrow 3q = 0 \Leftrightarrow q = 0.$$

$$103. \frac{x}{5} - \frac{x}{6} = 4 \Leftrightarrow \frac{6x - 5x}{30} = 4 \Leftrightarrow x = 120.$$

$$104. \frac{3x}{2y} = \frac{21}{22} \Rightarrow \frac{x}{y} = \left(\frac{21}{22} \times \frac{2}{3}\right) = \frac{7}{11} \Rightarrow x = \frac{7}{11}y.$$

$$4x + 5y = 83 \Rightarrow 4 \times \frac{7}{11}y + 5y = 83 \Rightarrow \frac{28}{11}y + 5y = 83$$

$$\Rightarrow 83y = 83 \times 11 \Rightarrow y = 11.$$

$$\therefore x = \frac{7}{11}y = \left(\frac{7}{11} \times 11\right) = 7.$$

$$\text{So, } y - x = 11 - 7 = 4.$$

$$105. \frac{ab}{a+b} = \frac{\frac{x}{x+y} \cdot \frac{y}{x-y}}{\frac{x}{x+y} + \frac{y}{x-y}} = \frac{\frac{xy}{x^2 - y^2}}{\frac{x(x-y) + y(x+y)}{x^2 - y^2}} = \frac{xy}{x^2 + y^2}.$$

$$106. \frac{a}{b} = \frac{1}{3} \Rightarrow b = 3a; \frac{b}{c} = 2 \Rightarrow c = \frac{b}{2} = \frac{3a}{2};$$

$$\frac{c}{d} = \frac{1}{2} \Rightarrow d = 2c = 2\left(\frac{3a}{2}\right) = 3a;$$

$$\frac{d}{e} = 3 \Rightarrow e = \frac{d}{3} = \left(\frac{3a}{3}\right) = a; \frac{e}{f} = \frac{1}{4} \Rightarrow f = 4e = 4a.$$

$$\therefore \frac{abc}{def} = \frac{(a)(3a)\left(\frac{3a}{2}\right)}{(3a)(a)(4a)} = \frac{9}{2}a^3 \times \frac{1}{12a^3} = \frac{3}{8}.$$

$$107. \frac{m}{n} = \frac{4}{3} \text{ and } \frac{r}{t} = \frac{9}{14} \Rightarrow \frac{mr}{nt} = \frac{4}{3} \times \frac{9}{14} = \frac{6}{7}.$$

$$\therefore \frac{3mr - nt}{4nt - 7mr} = \frac{3\frac{mr}{nt} - 1}{4 - 7\frac{mr}{nt}} = \frac{3 \times \frac{6}{7} - 1}{4 - 7 \times \frac{6}{7}}$$

$$= \frac{\frac{18}{7} - 1}{4 - 6} = \frac{11}{7} \times \left(-\frac{1}{2}\right) = -\frac{11}{14}.$$

$$108. a + \frac{1}{b} = 1 \Rightarrow ab + 1 = b \Rightarrow ab - b = -1$$

$$\Rightarrow b(a - 1) = -1 \Rightarrow b = \frac{1}{(1-a)}.$$

$$b + \frac{1}{c} = 1 \Rightarrow bc + 1 = c \Rightarrow bc - c = -1$$

$$\Rightarrow c(b - 1) = -1 \Rightarrow c = \frac{1}{(1-b)}.$$

$$\therefore c + \frac{1}{a} = \frac{1}{a} + \frac{1}{a} = \frac{1}{1 - \left(\frac{1}{1-a}\right)} + \frac{1}{a} = \frac{1}{\frac{(1-a)-1}{(1-a)}} + \frac{1}{a}$$

$$= \frac{(1-a)}{-a} + \frac{1}{a} = \frac{(a-1)}{a} + \frac{1}{a} = \frac{a-1+1}{a} = \frac{a}{a} = 1.$$

$$109. y + \frac{1}{z} = 1 \Rightarrow yz + 1 = z \Rightarrow yz - z = -1$$

$$\Rightarrow z(y - 1) = -1 \Rightarrow z = \frac{1}{(1-y)}.$$

$$x + \frac{1}{y} = 1 \Rightarrow xy + 1 = y \Rightarrow xy - y = -1$$

$$\Rightarrow y(x - 1) = -1 \Rightarrow y = \frac{1}{(1-x)}.$$

$$\therefore xyz = x \left(\frac{1}{1-x}\right) \left(\frac{1}{1-y}\right) = x \left(\frac{1}{1-x}\right) \left(\frac{1}{1 - \frac{1}{1-x}}\right)$$

$$= x \left(\frac{1}{1-x}\right) \left(\frac{1-x}{-x}\right) = -1.$$

$$110. \frac{x}{(2x+y+z)} = a \Rightarrow x = a(2x+y+z) \quad \dots(i)$$

$$\frac{y}{(x+2y+z)} = a \Rightarrow y = a(x+2y+z) \quad \dots(ii)$$

$$\frac{z}{(x+y+2z)} = a \Rightarrow z = a(x+y+2z) \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get: $x + y + z$

$$= a(4x + 4y + 4z) \Rightarrow a = \frac{x+y+z}{4(x+y+z)} = \frac{1}{4}.$$

$$111. \frac{a}{b+c} = K \Rightarrow a = K(b+c) \quad \dots(i)$$

$$\frac{b}{c+a} = K \Rightarrow b = K(c+a) \quad \dots(ii)$$

$$\frac{c}{a+b} = K \Rightarrow c = K(a+b) \quad \dots(iii)$$